# KEY SHEET

## PHYSICS

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **AD** | 2 | **CD** | 3 | **ABC** | 4 | **AC** | 5 | **ABD** | 6 | **ACDB** |
| 7 | **0.25** | 8 | **1.87 to 1.88** | 9 | **12** | 10 | **0** | 11 | **10** | 12 | **1** |
| 13 | **C** | 14 | **C** | 15 | **D** | 16 | **C** | 17 | **3** | 18 | **4** |
| 19 | **4** | | | | | | | | | | |

## CHEMISTRY

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 20 | **ABC** | 21 | **AC** | 22 | **AC** | 23 | **BCD** | 24 | **BCD** |
| 25 | **ABCD** | 26 | **0.21** | 27 | **8.3** | 28 | **9** | 29 | **0.07-0.08** |
| 30 | **4** | 31 | **10** | 32 | **D** | 33 | **C** | 34 | **C** |
| 35 | **A** | 36 | **3** | 37 | **5** | 38 | **7** | | |

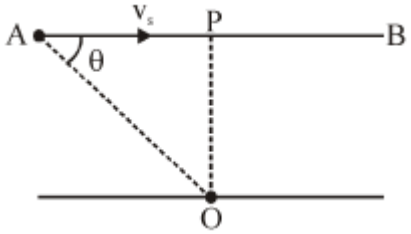
## MATHEMATICS

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 39 | **AC** | 40 | **C** | 41 | **ABD** | 42 | **BD** | 43 | **AD** |
| 44 | **ABD** | 45 | **0.20** | 46 | **2.20** | 47 | **121** | 48 | **484** |
| 49 | **222** | 50 | **0** | 51 | **D** | 52 | **B** | 53 | **A** |
| 54 | **D** | 55 | **2** | 56 | **8** | 57 | **22** | | |

# SOLUTIONS
# PHYSICS

1. (AD)
The wedges meet below the centre C, At a time, only one wedge that is farther away from the centre C slides relative to the bar.

2. (C,D)
The graph shows the situation shown in figure below. The observed frequency will initially be more than the natural frequency. When the source is at P, observed frequency is equal to its natural frequency i.e., 2000 Hz.




For region AP : $f = f_0\left(\frac{v}{v - v_s\cos\theta}\right)$ For PB : $f = f_0\left(\frac{v}{v + v_s\cos\theta}\right)$

Minimum value of f will be : $f_{\min} = f_0\left(\frac{v}{v + v_s}\right)$ when $\cos\theta = 1$ Or

$1800 = 2000\left(\frac{3000}{300 + v_s}\right)$, Solving this we get, $v_s = 33.33\ \text{m/s}$

And maximum value of f can be

$f_{\max} = f_0\left(\frac{v}{v - v_s}\right)$ when $\cos\theta = 1$ Or $f_{\max} = 2000\left(\frac{300}{300 - 33.33}\right) = 2250 Hz$

3. (A,B,C)
Light will emerge from prism if $A < \theta_C$.
Light will not emerge from prism if $A > 2\theta_C$ $\therefore$ $\theta_C = 45°$

4. (A,C)
Using $C = \frac{1}{\sqrt{\mu_0 \in_0}}$ and $R = \sqrt{\frac{\mu_0}{\in_0}}$ we can check the correctness.

A) $\mu_0 I^2 = \in_0 V^2$ $\frac{\mu_0}{\in_0} = \frac{V^2}{I_2} = R^2$ $\therefore$ $R^2 = R^2$ correct

B) $\in_0 I = \mu_0 V$ $\frac{\in_0}{\mu_0} = \frac{V}{I}$ $\frac{1}{R^2} = R$ not correct

C) $I = \in_0 cV$ $\frac{I}{V} = \in_0 c = \frac{\in_0}{\sqrt{\mu_0 \in_0}}$ $\frac{1}{R} = \sqrt{\frac{\in_0}{\mu_0}} = \frac{1}{R}$ correct

D) $\mu_0 CI = \in_0 V$ $\frac{\mu_0 C}{\in_0} = \frac{V}{I}$ $\frac{\mu_0}{\in_0}\frac{1}{\sqrt{\mu_0 \in_0}} = R$ $\therefore$ $\frac{1}{\in_0}\sqrt{\frac{\mu_0}{\in_0}} = R \Rightarrow \frac{R}{\in_0} = R$

incorrect.

5. (A,B,D) $E_{excitation} = \frac{1}{2}\frac{m}{2}u^2 = E_2 - E_1 = \frac{E_0}{2}$

$$\lambda_0 = \frac{hc}{\sqrt{2m_e\left[(E_2 - E_1) - \phi\right]}} = \frac{hc}{\sqrt{2m_e\left[\frac{E_0}{2} - \phi\right]}} = \frac{hc}{\sqrt{m_e E_0 - 2\sigma m_e}}$$

Also $\Rightarrow$ Energy transferred

$$\Rightarrow \frac{1}{2}mu^2 - \frac{1}{2}m\left(\frac{u}{2}\right)^2 = mu^2\left(\frac{1}{2} - \frac{1}{8}\right) = mu^2\left(\frac{3}{8}\right) = \frac{mu^2}{4}, \frac{3}{2} = \frac{E_0}{2}.\frac{3}{2} = -\frac{3E_0}{4}$$

Also For second case, $E_{excitation} = \frac{1}{2}\frac{4m}{5}u'^2 = E_2' - E_1' = 4(E_2 - E_1) = \frac{4}{5}E_0'$

$$E_0' = 5(E_2 - E_1) = \frac{5E_0}{2}z$$

6. (ABCD)

$\frac{dm}{dt} = 5kg/s$ $\quad v_C = 0.75\,m/s$

$F_{ext} - f = 0$ $\quad F_{ext} = f$

$\vec{f} = \frac{m d\vec{v}}{dt} - \vec{v}_{rel}\frac{dm}{dt}$ $\quad f\,\hat{i} = -\left(0 - v_C\hat{i}\right)\frac{dm}{dt}$ $\quad f = 0.75\times 5$ $\quad f = 3.75N$

$W = \int_i^f F_{ext}dS = 3.75(S) = 3.75(v_C t)$ $\quad W = 3.75(0.75\times 1)$ $\quad = 2.81J$

$\frac{dk}{dt} = \frac{1}{2}\left(\frac{dm}{dt}\right)v^2$ $\quad = \frac{1}{2}(5)\times(0.75)^2$ $\quad = 1.41J$ .

7. (0.25)

$P_0V_0^{5/3} = P_1(V_0 - \Delta V)P_0V_0$ $\quad \frac{P_0V_0}{T_0} = \frac{P_1(V_0 + \Delta V)}{T_0 + \Delta T}$

$\frac{P_0V_0^{2/3}}{T_0} = \frac{(V_0 - \Delta V)^{5/3}}{T_0 + \Delta V}$ $\quad 1 + \frac{\Delta T}{T_0} = \left(1 + \frac{\Delta V}{V_0}\right)\times\left(1 - \frac{\Delta V}{V_0}\right)^{-5/3}$ $\quad \frac{\Delta T}{T_0} = \frac{8\Delta V}{3V_0}$

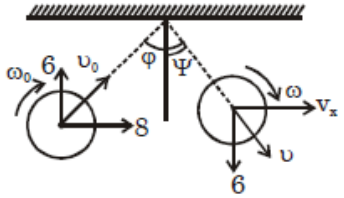
$T_0V_0V^{2/3} = (T_0 + \Delta T_2)(V_0 - \Delta V)^{2/3}$

$1 - \frac{\Delta T_2}{T_0} = \left(1 - \frac{\Delta V}{V_0}\right)^{2/3}$ $\quad \frac{\Delta T_2}{T_0} = \frac{2}{3}\frac{\Delta V}{V_0}$ $\quad \Rightarrow \Delta T_2\frac{\Delta T}{4} \Rightarrow Q = \frac{3}{2}R\Delta T + \frac{3}{2}R\Delta T_2 = \frac{15}{8}R\Delta T$

8. (1.87 to 1.88)
9. (12.00)
10. (0.00)

Sol. For (9 & 10)

Collision is elastic so the velocity perpendicular to wall will be is equal to 6 m/s after collision.



Impulse of normal $= m\left(v_f - v_i\right)_y$ $\quad I_N = -12\,m$

Impulse of friction $= \mu I_N = -1.2\,m$

Impulse momentum theorem along x-axis.

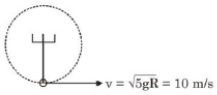
$I_N = \left(P_x\right)_f - \left(P_x\right)_i \quad -12\,m = m.v_x - m\times 8 \quad v_x = 6.8\,m/s \quad v = \sqrt{v_x^2 + 6^2} = 9.068$

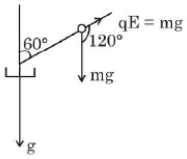
Angular impulse of friction about center of washer $J_f = -1.2\,m\times R$

Angular impulse angular momentum theorem $J_f = L_f - L_i$

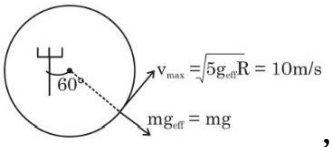
$$-1.2m\times R = L_f - \frac{mR^2}{2}(60) \qquad L_f = 0 \qquad \omega_f = 0$$

11. (10.00)
12. (1.00) for (11 & 12)




$$F_{net} = \sqrt{(mg)^2 + (mg)^2} + 2\times mg\times mg\times\frac{1}{2} = mg$$

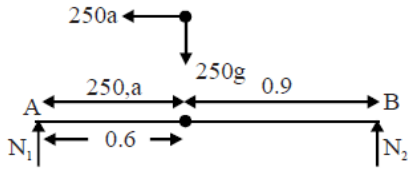



, Both are identical so $\frac{T_1}{T_2} = 1$.

13. (C) $\dfrac{Q^2}{192\pi\varepsilon_0 r}$

14. (C) Heat dissipated in $R_2$ is $\dfrac{Q^2}{24\pi\varepsilon_0 r}$ and no heat is dissipated in $R_1$.

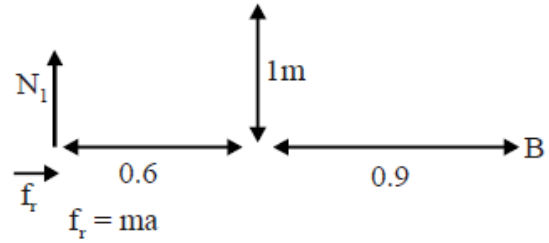
15. (D)




Balancing torque about B $\quad \Rightarrow N_1\times 1.5 = 250\,g\times 0.9 + 250a\times 2$

$N_1 = 5500/3N$

16. (C)




If $N_2 = 0 \Rightarrow N_1 = 2500$ N

Balancing torque about COM

$2500\times 0.6 = 250\times a\times 1m \quad \Rightarrow a = 6\,m/s^2$

17. (3) Apply Bernolli's equation b/w point P & D

$P_{atm} + \rho g h_1 + \rho V_p^2 = P_{atm} + \rho g(o) + \rho v_D^2$

[Assume zero level D]

$V_p \approx 0$ Area of cross section is very large $\quad V_D = \sqrt{2gh_1}$

Since area at C is half than area at D

so according to continuity equation

$$V_C A_C = V_D A_D \quad \Rightarrow \quad V_C = 2V_D = \sqrt{2gh_1}$$

Now for point P & C according to Bernolli's equation.

$$P_p + \rho g h_1 + \frac{1}{2}\rho V_p^2 = P_C + \rho g(0) + \frac{1}{2}\rho . V_C^2 \qquad V_p \approx 0$$

$$\Rightarrow \quad P_{atm} + \rho g h_1 = P_C + \frac{1}{2}\rho . (2\sqrt{2gh_1})^2$$

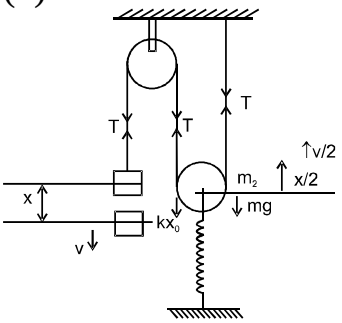
$$\Rightarrow \quad P_{atm} = P_C + 3\rho g h_1 \qquad .....(i)$$

for point E

$$P_E = P_C + \rho g h_2 = P_{atm} \qquad .....(ii)$$

from (i) & (ii) $\quad 3\rho g h_1 = \rho g h_2 \Rightarrow h_2 = 3h_1$

18. (4)




(i) Equilibrium position determination

$M_1 g = T \rightarrow$ (from FBD of $M_1$)

$2T = M_2 g + kx_0$ (from FBD of $M_2$)

$\therefore \quad 2M_1 g = M_2 g + kx_0 \qquad \therefore \quad kx_0 = 2M_1 g - M_2 g$

(ii) Displace block $M_1$ by small disp. x by

At new displaced position

$$-Mgx + \frac{1}{2}M_1 V^2 + \frac{1}{2}M_2\left(\frac{v}{2}\right)^2 + M_2 g\left(\frac{x}{2}\right) + \frac{1}{2}K\left(x_0 + \frac{x}{2}\right)^2 = C$$

Differentiating equation

$$-M_1 g\frac{dx}{dt} + \frac{1}{2}M_1 2v\frac{dv}{dt} + \frac{M_2}{8}2v\frac{dv}{dt} + \frac{M_2 g}{2}\frac{dx}{dt} + \frac{K}{2}2\left(x_0 + \frac{x}{2}\right)\left(\frac{1}{2}\frac{dx}{dt}\right) = 0$$

$$\Rightarrow -M_1 g + M_1 a + \frac{M_2 a}{4} + \frac{M_2 g}{2} + \frac{K}{2}\left(x_0 + \frac{x}{2}\right) = 0 \qquad \left(\text{where } a = \frac{dv}{dt}\right)$$

$$\Rightarrow -M_1 g + \frac{M_2 g}{2} + M_1 a + \frac{M_2 a}{4} + \frac{Kx_0}{2} + \frac{Kx}{4} = 0 \left(\text{from equilibrium } -M_1 g + \frac{M_2 g}{2} + \frac{Kx_0}{2} = 0\right)$$

Hence, $\frac{4M_1 + M_2}{4} a = \frac{-Kx}{4} \quad \therefore \quad a = -\left(\frac{K}{4M_1 + M_2}\right)x \quad \omega^2 = \frac{K}{(4M_1 + M_2)}$

$$\omega = \sqrt{\frac{K}{(4M_1 + M_2)}} \; ; \; T = \frac{2\pi}{\omega} \quad \therefore \quad T_2 = 2\pi\sqrt{\frac{4M_1 + M_2}{K}}$$

19. (4)

# CHEMISTRY

20. $P_{ext}=0 \qquad q=0 \qquad \Delta S_{sys}=nC_v\ell n\dfrac{V_2}{V_1}$

$W=-P_{ext}\Delta V \qquad \Delta U=0 \qquad T_1=T_2$

$W=0 \qquad \Delta H=0$

$T_1=T_2 \qquad \Delta S_{surrounding}=0 \; \Delta S_{sys}=nR\ell n\dfrac{V_2}{V_1}$

21. $\underset{5-x}{PCl_5(g)} \rightleftharpoons \underset{x}{PCl_3(g)}+\underset{x}{Cl_2(g)}$

$m_r=4+5-x+x+x \quad =9+x$

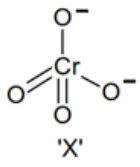
$K_p=\dfrac{\frac{3}{12}\times 4.8\times\frac{3}{12}\times 4.8}{\frac{2}{12}\times 4.8}=\dfrac{3}{12}\times\dfrac{4.8\times 3}{2} \qquad =1.8atm$

$PV=nRT \qquad n=\dfrac{4.8\times 49.2}{0.082\times 240}$

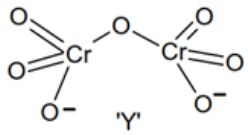
$9+x=12 \qquad x=12-9 \qquad x=3 \qquad \alpha=\dfrac{3}{5}=0.6$

22. $FeCr_2O_4+Na_2CO_3+O_2\rightarrow \underset{X}{Na_2CrO_4}+Fe_2O_3+CO_2$

$Na_2CrO_4+H^+\rightarrow \underset{'Y'}{Na_2Cr_2O_7}+H_2O$



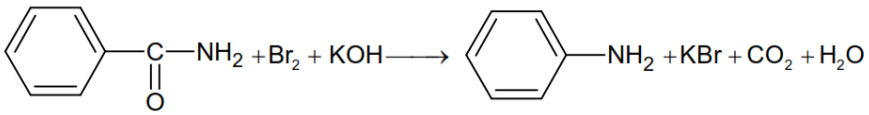
'X'



'Y'

23. $Cu+conc.HNO_3\rightarrow \underset{'K'}{Cu(NO_3)_2}+\underset{'L'}{NO_2}\uparrow+\underset{'M'}{H_2O}$

$Cu(NO_3)_2+KCN(excess)\rightarrow \underset{'T'}{K_3}\left[Cu(CN)_4\right]+\underset{'S'}{(CN)_2}\uparrow+KNO_3$

24. 


25. $-NH_2$ is $1^0$ amine and in rest of option lone pair of star (*) marked nitrogen is taking part in resonance. Or $sp^2$ hybridized.

26&27. At phenolphthalein end point $\dfrac{X}{106}=5\times 20\times 0.01\times 10^{-3}$

At methylorange end point $\dfrac{X}{53}+\dfrac{Y}{84}=5\times 60\times 0.01\times 10^{-3}$

At phenolphthalein end point the solution is of only $NaHCO_3 \qquad \therefore pH=\dfrac{pK_{a_1}+pK_{a_2}}{2}$

28&29. This is a case of parallel reaction,

$k_1+k_2=\dfrac{2.303}{t}\log\left(\dfrac{a}{a-x}\right)$

Where, $t=20\min, x=0.80a \; \therefore k_1+k_2=\dfrac{2.303}{20}\log\dfrac{a}{0.20a} \qquad =0.08048\min^{-1}$

At this moment

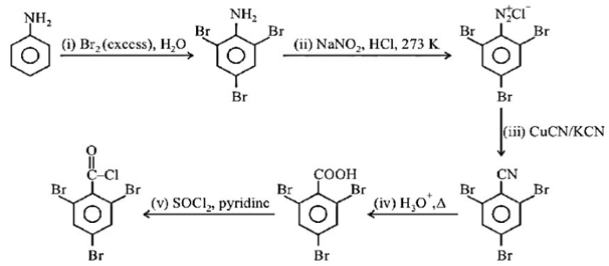
$[B]_t = 0.90a, [A]_t = 0.10a$

$[C]_t = 0.10a, [B]_t = 0.90a \therefore \frac{d[B]}{dt} = k_1[A]_t = k_1 \times 0.10a$ and $\frac{d[C]}{dt} = k_2 \times 0.90a$

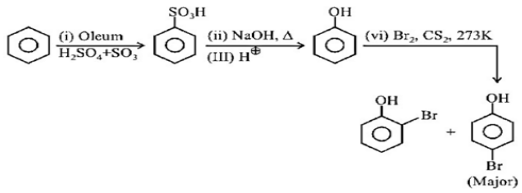
Thus, $k_1 \times 0.10a = k_2 \times 0.90a \therefore \frac{k_1}{k_2} = 9 \Rightarrow k_1 = 9k_2 \therefore k_2 = \frac{k_1}{9}$

$k_1 + \frac{k_1}{9} = 0.08048\,\text{min}^{-1}$ $\frac{10k_1}{9} = 0.08048\,\text{min}^{-1}$ $\therefore k_1 = 0.08048 \times \frac{9}{10} = 0.0724\,\text{min}^{-1}$

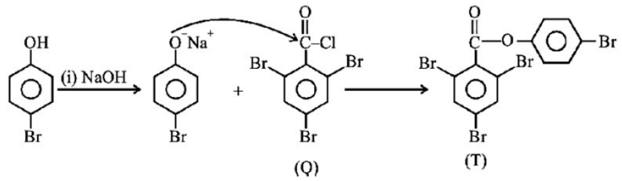
30 Scheme:1

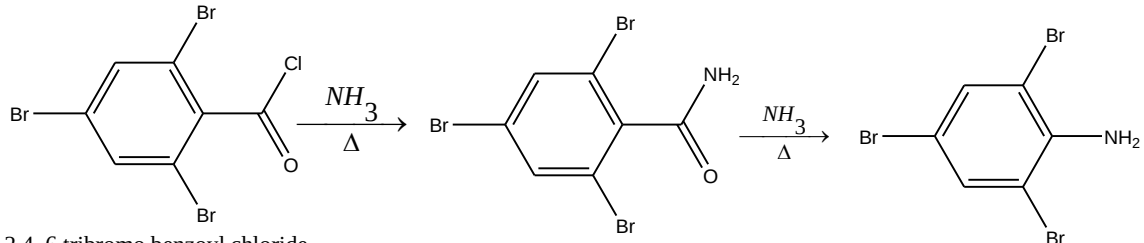



Scheme:2




Scheme:3

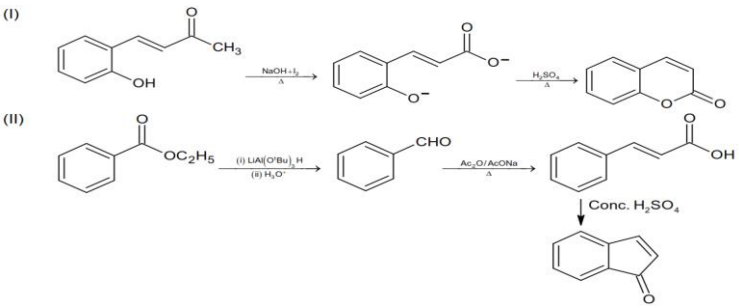


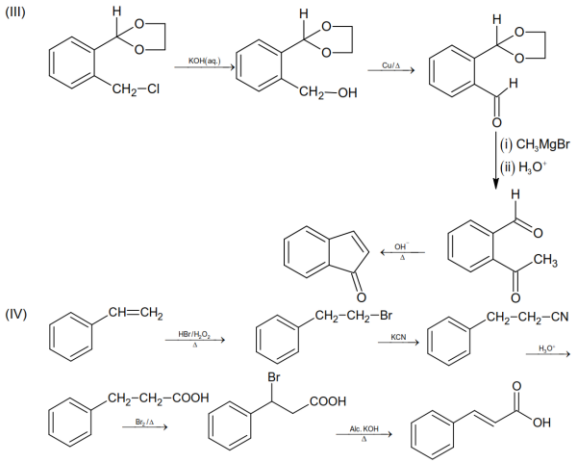




2,4, 6 tribromo benzoyl chloride — 2,4, 6 tribromo benzamide — 2,4, 6 tribromo aniline

31.

32&33.






34&35. $FeCl_3 \rightarrow$ Yellow solution and does not react with $dil\ H_2SO_4$

$2Fe^{3+} + H_2S \xrightarrow{H^+} 2\underset{U}{Fe^{2+}} + S$

$Fe^{3+} + 3NaOH \underset{(excess)}{\rightarrow} \underset{Brown\ ppt(V)}{\left(Fe(OH)_3\right)\downarrow} + 3Na^+$

36. $Cu^{2+} + 2e^- \rightarrow Cu \qquad H_2O \rightarrow 2H^+ + \frac{1}{2}O_2 + 2e^-$

$Q = 9.65 \times 10 = 96.5C$

Moles of $H_2SO_4$ produced = moles of $Cu^{2+}$ reduced $= \frac{96.5}{2 \times 96500} = 5.0 \times 10^{-4}$

Conc. Of $H_2SO_4\,(M) = \frac{5 \times 10^{-4}}{1} = 5 \times 10^{-4}\ mole/L$

$\left[H^+\right] = 2 \times 5 \times 10^{-4} = 10^{-3}\ mole/L$

$pH = -\log\left[H^+\right] = 3$

37. $(NH_4)_2\,SO_4 + 2NaOH \rightarrow 2NH_3 + Na_2SO_4 + 2H_2O$

$n = 2.5$

$\underset{n=2}{CuSO_4.5H_2O} + \underset{\substack{n=5 \\ \text{limiting}}}{4NH_3} \rightarrow \left[Cu\left(NH_3\right)_4\right]SO_4 + 5H_2O$

Hence, moles of $\left[Cu\left(NH_3\right)_4\right]SO_4 = \frac{5}{4} = 1.25$

38. $3Cl_2 + 6NaOH\,(conc.) \rightarrow 5NaCl + NaclO_3 + H_2O$

$3Br_2 + 6NaOH\,(conc.) \rightarrow 5NaBr + NaBrO_3 + 3H_2O$

$3l_2 + 6NaOH\,(conc.) \rightarrow 5Nal + NalO_3 + 3H_2O$

$3H_2O + P_4 + Conc.3NaOH \rightarrow PH_3 + 3NaH_2PO_2$

$(CH_3)_2\,CHCHO + Conc.NaOH \rightarrow (CH_3)_2\,CHCOO^-Na^+ + (CH_3)_2\,CHCH_2OH$

$HCHO + Conc.NaOH \Rightarrow HCOO^-Na^+CH_3OH$

Total 6 compound undergo disproportionation.

So $x = 6$ & $y = 5$ $x/y = 1.2$

# MATHEMATICS

39. $S_n = \sum_{r=1}^{n} T_r = n\left(2n^2+9n+13\right) \Rightarrow T_r = S_r - S_{r-1} \quad f\left(\sqrt{2}\right) = \sqrt{3}\left(\sqrt{2}+3\right)$

$= r\left(2r^2+9r+13\right) - (r-1)\left(2(r-1)^2+9(r-1)+13\right)$

$= 6r^2+12r+6 = 6(r+1)^2 \quad \Rightarrow \sqrt{T_r} = \sqrt{6}(r+1)$

$\Rightarrow \sum_{r=1}^{n} \sqrt{T_r} = \sqrt{6}\sum_{r=1}^{n}(r+1) \quad = \sqrt{6}\left(\frac{n^2+3n}{2}\right) \quad = \sqrt{\frac{3}{2}}\left(n^2+3n\right)$

Also, $\sqrt{2}$ is not in the domain

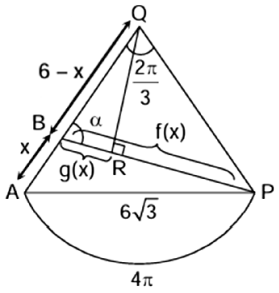
40. $I = \int_{\pi}^{2\pi} f^{-1}(x)\,dx$

Putting $x = f(t)$ or $dx = f'(t)\,dt$, we get

$I = \int_{\pi}^{2\pi} t.f'\,dt$

$= 2\pi f(2\pi) - \pi f(\pi) - \left|\frac{t^2}{2} - \cos t\right|_{\pi}^{2\pi} = \frac{3\pi^2}{2} + 2$

41. In $\Delta PQB \quad \cos\frac{2\pi}{3} = \frac{6^2+(6-x)^2-\left(f(x)\right)^2}{12(6-x)} = -\frac{1}{2}$




$\Rightarrow f(x) = \sqrt{x^2-18x+108} \forall x \in [0,6]$

In $\Delta PQB \quad \cos\alpha = \frac{g(x)}{6-x} = \frac{(6-x)^2+\left(f(x)\right)^2-6^2}{2f(x).(6-x)} \quad \Rightarrow g(x) = \frac{x^2-15x+54}{\sqrt{x^2-18x+108}}$

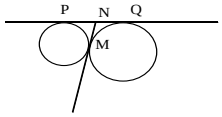
R is a point from where down-hill road will begin

$\Rightarrow g(x)$ will be maximum where x=0 i.e. when QR is maximum

$\Rightarrow g(0) = \frac{54}{\sqrt{2.54}} = 3\sqrt{3}$ *and* $f(0) = 6\sqrt{3}$

42. MN is radical axis and MN = NP = NQ

$\therefore$ locus of M is circle with centre N = $N = \frac{P+Q}{2}$ and $r = NQ \text{ or } NP = \sqrt{40}$




Locus of M is circle centre (0,1) and radius $\sqrt{40}$, excluding points P and Q

$\Rightarrow x^2+y^2-2y = 39$ excluding points P and Q

$\therefore (-2,7)$ is excluded from locus and $\left(0,\frac{3}{2}\right)$ does not lie in $E_1$

Locus of point in $E_2$ is circle with diameter (0,1) and (1,1), excluding midpoints of chords passing through P or Q $\Rightarrow x^2-x+(y-1)^2=0$

$\therefore\left(\frac{1}{2},1\right)$ does not lie in $E_2$ and $\left(\frac{4}{5},\frac{7}{5}\right)$ doesn't lie in $E_2$ since latter is midpoint of chord through P

43. $\Delta=\begin{vmatrix}-1 & 2 & 5\\ 2 & -4 & 3\\ 1 & -2 & 2\end{vmatrix}=0$ given equation represent family of lines

$\Rightarrow 2x-4y+3z-b_2+\lambda(-x+2y+5z-b_1)=0$ is same as $x-2y+2z-b_3=0$ for same $\lambda$

$\frac{2-\lambda}{1}=\frac{2\lambda-4}{-2}=\frac{5\lambda+3}{2}=\frac{-(b_1\lambda+b_2)}{-b_3}$ $\lambda=\frac{1}{7}$ $and\, 13b_3=b_1+7b_2$ $A\rightarrow\begin{vmatrix}1 & 2 & 3\\ 0 & 4 & 5\\ 1 & 2 & 6\end{vmatrix}\neq 0$

$B\rightarrow\begin{vmatrix}1 & 1 & 3\\ 5 & 2 & 6\\ 2 & 1 & 3\end{vmatrix}\neq 0=0$ as planes are non parallel is must represent family of planes for solution to exit $(5\mu+1)x+(2\mu+1)y+(6\mu+3)z=b_1+\mu b_2$

Must be $2x+b+3z+b_3=0$ which is not true always

$C\rightarrow x-2y+5z=-b_1, x-2y+5z=\frac{b_2}{2}, x-2y+5z=b_3$ as these are parallel for specific case

when $-b_1=\frac{b_2}{2}=b_3$ $D\rightarrow\begin{vmatrix}1 & 2 & 5\\ 2 & 0 & 3\\ 1 & 4 & -5\end{vmatrix}\neq 0$

44. The line should be either coincident on $P_1$ or on $P_2$ or interest on $P_1$ and $P_2$ on different points.

$D)\frac{x}{1}=\frac{y-4}{-2}=\frac{z}{3}=\lambda$ $\Rightarrow(\lambda,-2\lambda+4,3\lambda)$ lie on $P_2$

$A)\frac{x-1}{0}=\frac{y-1}{0}=\frac{z-1}{5}$ intersects $P_1$ and $P_2$ on different points.

$B)\frac{x-6}{5}=\frac{y}{2}=\frac{z}{3}$ also intersects $P_1\, and\, P_2$ on different points.

45.

| | $S_1$ | $F_2=F_1\cup S_1$ | $S_2$ | $G_1\cup S_2=G_2$ | $S_3$ |
|---|---|---|---|---|---|
| i) | $\{1,2\}$ | $\{1,2,3,4\}$ | $\{1,x\}$ | $\{1,2,3,4,5\}$ | $\{1,2\}$ |
| ii) | $\{2,3\}$ | $\{1,2,3,4\}$ | $\{1,x\}$ | $\{1,2,3,4,5\}$ | $\{2,3\}$ |
| | | $\{x,y\}$ (where x and y) | Or $\{2,3,4,5\}$ | | |
| iii) | $\{1,3\}$ | $\{1,3,4\}$ | $\{1,x\}$ are other than 1 | $\{1,2,3,4,5\}$ | $\{1,3\}$ |

i) $P_1=\frac{{}^2C_2}{{}^3C_2}\cdot\frac{{}^3C_1}{{}^4C_2}\cdot\frac{{}^2C_1}{{}^5C_2}=\frac{1}{60}$

ii) $P_2 = \frac{1}{3}\left(\frac{{}^3C_1}{{}^4C_2}.\frac{{}^2C_2}{{}^5C_2}.\frac{{}^3C_2}{{}^4C_2}\times\frac{{}^2C_2}{{}^4C_2}\right) = \frac{2}{45}$

iii) $P_3 = \frac{1}{{}^3C_1}\times\frac{{}^2C_1}{{}^3C_2}\times\frac{{}^2C_2}{{}^5C_2} = \frac{1}{45}$

Conditional probability $= \frac{P_1}{P_1+P_2+P_3} = \frac{1}{5}$

46. As in the previous table, $F_2 = \{1,2,3,4\}$ when $S_1 = \{1,2\}\, or\, \{2,3\}$

$q = (18+48)/(18+48+24)$

47. $A = \begin{bmatrix} a & b \\ c & a \end{bmatrix}$ where a, b, c $\in \{0,1,2,.....p-1\}$

Case-1 A is symmetric matrix $\Rightarrow b = c$

$\Rightarrow \det(A) = a^2 - b^2$ is divisible by p $\Rightarrow (a-b)(a+b)$ is divisible by p

a) $a-b$ is divisible by p if $a = b,$ then 'p' cases are possible

b) $a+b$ is divisible by p if $a+b = p,$ then $\frac{(p-1)}{2}\times 2 = (p-1)$ cases are possible

Case-II

A is skew symmetric matrix

If $a = 0, b+c = 0,$ then $\det(A) = b^2$

$\Rightarrow b^2$ can never be divisible by p. So no case is possible

Total number of A is possible $= 2p-1$

48. The trace is '2a' which is divisible by p if and only if $a = 0$. So we assume $a \neq 0$. For each such fixed 'a' we now need to determine all ordered pairs $(b,c)$ for which $a^2 - bc$ is divisible by p. Note that even through $1 \le a \le p-1, a^2$ may be bigger than p. Let r be the remainder when $a^2$ is divided by p. Then $r \neq 0$ (as otherwise p would divide a) and so $1 \le r \le p-1$. The condition that $a^2 - bc$ is divisible by p is equivalent to saying that the integer 'bc' also leaves the remainder r when divided by p. This rules out the possibility that $b = 0$. For every b with $1 \le b \le p-1,$ we claim that there is precisely one c with $1 \le c \le p-1,$ such that 'bc' leaves the remainder r when divided by p. To see this, consider the remainders when the multiples $b, 2b, 3b, ....(p-1)b$ are divided by p. None of these remainders is 0 as otherwise p would divide b. We claim these remainders are all distinct. For, suppose ib and jb leave the same remainder for some i,j with $1 \le i \le p-1$. Then p divides $(j-i)b$ which would mean that either p divides $j-i$ or it divides b, both of which are impossible since $j-i$ and b both lie between 1 and p-1. Since the remainders left by b, 2b,...(p-1)b are all distinct, and there are p-1 possible values for the remainder, we conclude that the remainder r occurs precisely once in this list. Put differently, for each $b \in \{1,2,.., p-1\}$, there is precisely one c such that 'bc' leaves the same remainder as $a^2$ does, which is equivalent to saying that $a^2 - bc$ is divisible by p. So, for each fixed $a \in \{1,2,...., p-1\}$ there are $p-1$ matrices of the desired type. Since a itself takes $p-1$ distinct values, the total number of desired matrices is $(p-1)^2$.

49. It can be proven that $a^2+b^2+c^2 = 3\left(PA^2+PB^2+PC^2\right) - 9PG^2$

Let P be origin and position vectors of point A, B,C be $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$

$\left|\vec{\alpha}+\vec{\beta}+\vec{\gamma}\right|^2+\left|\vec{\alpha}-\vec{\beta}\right|^2+\left|\vec{\beta}-\vec{\gamma}\right|^2+\left|\vec{\gamma}-\vec{\alpha}\right|^2=3\left(\left|\vec{\alpha}\right|+\left|\vec{\beta}\right|+\left|\vec{\gamma}\right|^2\right)$

$\Rightarrow 9PG^2+AB^2+BC^2+CA^2=3\left(PA^2+PB^2+PC^2\right)$

50. Since $a^2+b^2+c^2$ is constant, $QA^2+QB^2+QC^2$ is minimum when Q is at G.

51.& 52. Let degree of $f(x)=n$

$\therefore$ Degree of $f'(x)=n-1$, degree of $f''(x)=n-2$

Since, $f''(x).f'(x)=f(x)$

$\Rightarrow(n-1)+(n-2)=n\Rightarrow n=3$

Let $f(x)=ax^3+bx^2+cx+d$

$\Rightarrow(6ax+2b)\left(3ax^2+2bx+c\right)=\left(ax^3+bx^2+cx+d\right)$

Comparing gives a = 1/18

53 & 54. Let $Z_1$ (purely imaginary) be a root of the given equation.

Then, $Z_1=-\bar{Z}_1$ and $az_1^2+bz_1+c=0.....(1)$

$\Rightarrow \bar{a}\bar{z}_1^2+\bar{b}\bar{z}_1+\bar{c}=0$ $\Rightarrow \bar{a}\bar{z}_1^2-\bar{b}\bar{z}_1+\bar{c}=0\left(as\ \bar{z}_1=-\bar{z}_1\right).....(2)$

Now eqns. (1) and (2)must have one common root.

$\therefore\left(c\bar{a}-a\bar{c}\right)^2=\left(b\bar{c}+c\bar{b}\right)\left(-a\bar{b}-\bar{a}b\right)$

Let $z_1$ *and* $z_2$ be two purely imaginary roots. Then, $\bar{z}_1=-z_1,\bar{z}_2=-z_2$

Now, $az^2+bz+c=0$ .....(1)

or $\overline{az^2+bz+c}=\bar{0}$ or $\bar{a}\bar{z}^2+\bar{b}\bar{z}+\bar{c}=0$

or $\bar{a}z^2+\bar{b}z+\bar{c}=0$.....(2)

Equation (1) and (2) must be identical as their roots are same.

$\therefore\frac{a}{a}=-\frac{b}{b}=\frac{c}{c}$ $\Rightarrow a\bar{c}=\bar{a}c, a\bar{b}+\bar{a}b=0$ *and* $b\bar{c}+\bar{b}c=0$

Hence, $a\bar{c}$ is purely real and $a\bar{b}$ and $b\bar{c}$ are purely imaginary.

55. Equation of normal at $\left(t^2,2t\right)$ is $y=-xt+2t+t^3$

Let any point on $y=2$ is $(h,2)$

If normal passes through (h,2)

$2+ht-2t-t^3=0$ $t^3-t(h-2)-2=0$

$t_1,t_2$ *and* $t_3$ are points of this equation vertices lies on $xy=-2$

This is because tangents at $t_1$ *and* $t_2$ meet at $\left(t_1,t_2,t_1+t_2\right)=\left(2/t_3,-t_3\right)$

56. $f(a,b)$ represent distance between $\left(\sqrt{25-a^2},a\right)$ *and* $\left(b,\frac{1}{3}\sqrt{144-16b^2}\right)$ which represent the curves $16x^2+9y^2=144$ *and* $x^2+y^2=25.$

So, $c=9$ *and* $d=1$

57. Let $p=9-a_2$ & $q=9-a_4$, then $p+a_1+q+a_3=9\Rightarrow {}^{12}C_3=220$